[LinkedIn]

Ian Wright

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[Website]

Education

University of Maryland, Baltimore County (UMBC) Baltimore, MD | Aug 2023 – May 2027

Bachelor of Science in Mechanical Engineering with a Minor in Mathematics with a GPA of 3.3/4.0

Related Coursework: Differential Equations, Fundamentals of Circuits and Electronics, Advance MATLAB, Fluid Dynamics, Solids and Materials Lab, Fundamentals of Machine Design, Advance Python, Dynamics, Control Systems

Skills

- **Mechanical & Analysis:** Mechanical design, Rapid Prototyping, Circuit Analysis, Geometric Dimensioning and Tolerancing (GD&T), Finite element Analysis (FEA), Materials Science
- **Modeling & Simulation:** SOLIDWORKS (CSWA), Fusion 360, Simulink, LT Spice, state-space modeling
- **Programming:** MATLAB, Python, C++ (Arduino), C
- **Manufacturing & Prototyping**: 3D printing, Lathe Machining, Soldering
- **Tools/Reporting:** Data visualization, technical reporting, quad charts, LaTeX

Experience

Materials Science Lab (ENME332L) | Teaching Fellow Baltimore, MD | Feb 2026 – Present

- Facilitating a Lab Section of over 20 students explaining experimental methods while reinforcing materials science concepts
- Emphasize defined and documented laboratory processes and safety protocols and helped with lab report writing

Design Build Fly UMBC AIAA| Project Intern Baltimore, MD | Nov 2025 – Present

- Led and taught a team of three to laminate and construct composite landing-gear structures using composite layup techniques, vacuum bagging, adhesive bonding, and proper curing practices to achieve competition-grade hardware.
- Designed and manufactured tangible, competition-ready prototypes for multiple fixed-wing aircraft configurations by applying, tolerance awareness, structural testing, and iterative prototyping.
- Performed precision soldering, wiring-harness routing, and electrical integration by cutting wire extensions to spec, applying heat-shrink tubing, and following assembly safety protocols and quality-control checks.

Estimated Control, and Learning Lab (ECLL) at UMBC| Research Assistant Baltimore, MD | Apr 2025 – Present

- Developing a dynamic stabilization system for an inverted pendulum that counterbalances itself using a reaction wheel actuator, focusing on real-time control, torque management, and system response
- Derived the equations of motion, converted them into state-space form, and desinging PID and state-feedback controllers in MATLAB/Simulink
- Designing and building a functional SOLIDWORKS assembly for the pendulum and reaction-wheel system, including motor selection, manufacturability considerations, and wire considerations
- Documenting the equations of motions in LaTeX, maintaining and updating the Bill of Materials (BOM), and documenting algorithm performance and design iterations for future autonomous-systems research.

Aberdeen Proving Ground Lab Fellowship| Test Engineer Intern Edgewood, MD | Jun 2025 – Aug 2025

- Worked with government and industry stakeholders to manage the development and testing of decontamination technologies for the Department of War.
- Created a SharePoint Directory to connect over 100 stakeholders allowing easy access to documents while maintaining control
- Designed an OSHA-compliant interactive test plan, simulating the use of a barrier material for fixed-site remediation using current military technologies.
- Presented an audiovisual overview of my accomplishments to the director’s level of agency leadership.

UAS Research and Operations Center (UROC) | R&D Engineer Intern California, MD | Jun 2024 – Aug 2024

- Developed, implemented and successfully demonstrated a net gun system for a “defender” drone to neutralize an “attacker” drone as a form of counter UAV.
- Designed the net gun solution using Fusion 360 and sourced components to maximize the use of common industry standards while also simulating a capture rate of 90%.

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- Assembled drones and applied soldering techniques while gaining awareness of NAVAIR programs and initiatives such as the ARTEMIS program.
- Delivered bi-weekly quad-chart presentation reporting to leadership about progress being made

Projects

Arduino/Raspberry Pi Light Control | UMBC — Joppa, MD | Jan 2026 – Feb 2026

- The goal was to control LED lights through both Arduino and Raspberry Pi while comparing them
- For the Arduino, it gave me easy control over all the pins with simple but effective software
- For the Raspberry Pi, I can remotely access using Secure Shell (SSH), but there is no native software
- Controlled the LED lights using both controllers counting the sequence and changing the speed with a potentiometer concluding that the Arduino is cost effective and simple to use where the Raspberry Pi has more potential but pricier

Analyzing Buckling and Compressive Failure of 3D Specimens | UMBC — Baltimore, MD | Aug 2025 – Dec 2025

- Design and execute an experiment in testing 24 additively manufactured Polylactic acid (PLA) specimens to evaluate mechanical performance under controlled loading conditions
- There were 24 total specimens tested across 4 infill densities (10%, 25%, 50%, 100%) and 2 lengths (77.5 mm and 125 mm)
- Wrote a lab handout and lab report consisting of over 20 pages and determined that at lower infills the specimens experienced compressive behavior and at higher infills the specimens experienced buckled behavior.
- Authored a 20+ page lab handout and lab report, determining that lower infill densities produced primarily compressive deformation while higher infill densities resulted in buckling-dominated failure

MATLAB Auto-Regressive Model of a Mass Spring System | UMBC — Baltimore, MD | Aug 2025 – Dec 2025

- Built MATLAB simulations of dynamic systems (mass-spring-damper) to simulate two different original datasets
- Benchmarking Eulers, Discretization, ODE45, and Simulink methods
- Evaluated accuracy and error and presented findings in a technical report lasting over 10 pages, finding Eulers method as the least accurate and Simulink as the most accurate with an error close to 0 for both datasets.

Lift Platform Challenge | UMBC — Baltimore, MD | Aug 2025 – Dec 2025

- Tasked with designing an 8”x8”x 8” device capable of lifting a 3.1lb weight 4” above its original height.
- Calculated Stress, Deflection, Failure modes by FEA and by hand for Drivetrain and platforming system.
- Created a functional CAD Assembly in SOLIDWORKS following GD&T practices.
- Designed a working switch to seamlessly control the Brushless DC motor through soldered wire connections.
- Participated in open forum discussions collecting feedback and documented all our work in multiple design updates cumulating in a 70+ page Preliminary Design Report (PDR)

Project AdaptAble | UMBC — Baltimore, MD | Jan 2025 – May 2025

- Designed an assistive-technology toy for children with limited hand mobility and visual impairments, focusing on usability and human-centered design.
- Used a “toy robot” concept to improve user engagement and create an interactive mechanical system.
- Built the design using a combination of 3D-printed components and commercial mechanical parts, incorporating rotational joints at the hips and shoulders along with magnetic attachment points for modularity.
- Documented the full design process in meeting the projects deliverables, presented the work in an open-forum Q&A session, and created a final engineering report summarizing prototypes, testing, and design decisions.

UMBC Robotics on the Surface (UROS) | UMBC — Baltimore, MD | Aug 2024 – Dec 2024

- Collaborated with a team to create a robot that will go through a course, retrieve objects (ping pong balls) and return them to pre-determined coordinates.
- Programmed controls through Robot C, and maximizing route efficiency within budgetary constraints collecting over 400 points.
- Presented challenges and mechanisms of our design to university robotics leaders.

Certifications and Associations

- SOLIDWORKS Certified Associate (CSWA)
- American Society of Professional Engineers (ASME) (Member)
- American Institute of Aeronautics and Astronautics (AIAA) (Project Intern)